

SMART WORKOUT SUGGESTION EXPERT SYSTEM



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Smart Workout Suggestion System

System Design Document

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1. Introduction

1.1. Purpose

The Smart Workout Suggestion System is a fuzzy logic-based expert system designed to provide personalized fitness and dietary recommendations. By analyzing a user's physiological data, lifestyle habits, and current fitness levels, the system aims to emulate the decision-making process of a professional fitness coach. Its primary purpose is to generate actionable advice on workout intensity and dietary adjustments to help users achieve their health and fitness goals in a safe and effective manner.

1.2. Scope

The system accepts seven key inputs: Age, Body Mass Index (BMI), Daily Activity Level, Resting Heart Rate, Strength Level, Flexibility, and Sleep Quality. Based on these inputs, it produces three outputs: an assessment of Overall Fitness, a recommended Workout Intensity, and a suitable Diet Recommendation. This document provides a comprehensive technical overview of the system's architecture, including its fuzzy inference engine, variable definitions, and the complete rule base that governs its logic.

1.3. System Overview

This system is implemented as a Mamdani-type Fuzzy Inference System (FIS), a model known for its intuitive, human-readable rule structure. It comprises 7 input variables, 3 output variables, and a knowledge base of 343 rules. The system processes crisp numerical inputs, converts them into fuzzy sets (fuzzification), applies the rule base to infer fuzzy outputs (inference), and then converts these fuzzy outputs back into crisp, actionable values (defuzzification).

2. System Architecture

2.1. Fuzzy Inference Engine

The core of the system is a Mamdani fuzzy inference engine configured with the following standard operators:

Parameter	Method	Description
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Fuzzy Logic Type	mamdani	A widely used inference method that produces fuzzy outputs.
AND Method	min	The fuzzy AND operator is implemented as the minimum of the membership values.
OR Method	max	The fuzzy OR operator is implemented as the maximum of the membership values.
Implication Method	min	The implication (IF-THEN) is modeled by truncating the output fuzzy set at the rule's firing strength.
Aggregation Method	max	The outputs of all rules are combined using the maximum of their fuzzy sets.
Defuzzification Method	centroid	The final crisp output value is calculated as the center of gravity of the aggregated fuzzy set.

2.2. Input Variables

The system utilizes seven input variables to assess the user's condition. All membership functions are triangular (trimf), defined by a vector $[a, b, c]$ where a and c are the "feet" of the triangle and b is the peak.

2.2.1. Age

The user's age in years. Range: [15, 70].

MF Name	Type	Parameters
Young	trimf	[15, 25, 35]
Adult	trimf	[30, 45, 60]
Old	trimf	[55, 65, 70]

2.2.2. BMI (Body Mass Index)

The user's BMI value. Range: [15, 40].

MF Name	Type	Parameters
Underweight	trimf	[15, 16.5, 18.5]
Normal	trimf	[18.5, 21.5, 25]
Overweight	trimf	[25, 27.5, 30]
Obese	trimf	[30, 32.5, 40]

2.2.3. Daily Activity Level

A categorical rating of the user's daily physical activity. Range: [1, 4].

MF Name	Type	Parameters
Sedentary	trimf	[1, 1.5, 2]
Lightly Active	trimf	[1.5, 2, 2.5]
Moderately Active	trimf	[2, 2.5, 3]
Very Active	trimf	[2.5, 3, 4]

2.2.4. Resting Heart Rate

The user's resting heart rate in beats per minute (BPM). Range: [40, 100].

MF Name	Type	Parameters
Low	trimf	[40, 50, 60]

Normal	trimf	[60, 70, 80]
High	trimf	[80, 90, 100]

2.2.5. Strength Level

A self-assessed rating of muscular strength. Range: [1, 10].

MF Name	Type	Parameters
Weak	trimf	[1, 2, 3]
Average	trimf	[3, 5, 7]
Strong	trimf	[7, 8.5, 10]

2.2.6. Flexibility

A self-assessed rating of physical flexibility. Range: [1, 10].

MF Name	Type	Parameters
Poor	trimf	[1, 2, 3]
Average	trimf	[3, 5, 7]
Good	trimf	[7, 8.5, 10]

2.2.7. Sleep Quality

A rating of recent sleep quality. Range: [0, 9].

MF Name	Type	Parameters
Poor	trimf	[0, 2.5, 5]
Average	trimf	[5, 6, 7]
Good	trimf	[7, 8, 9]

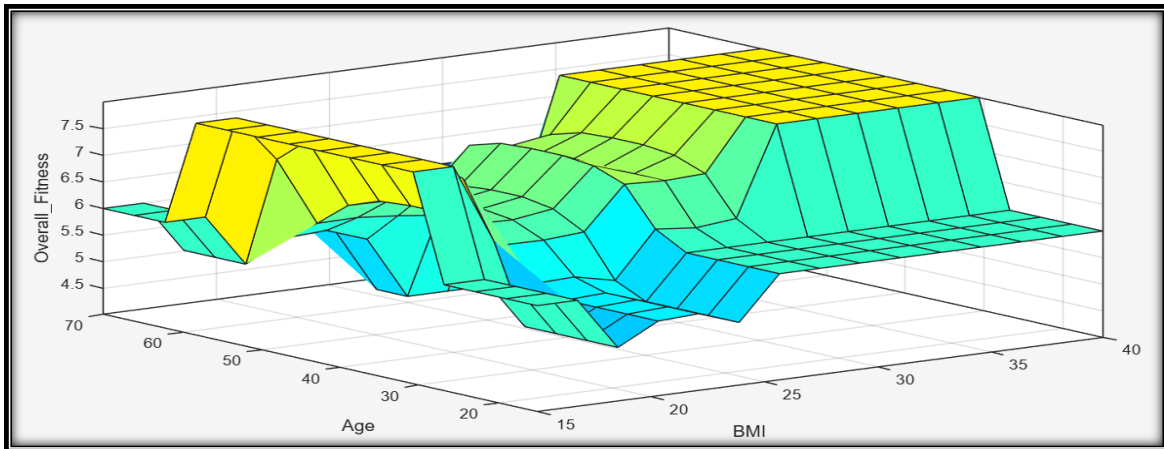
2.3. Output Variables

The system generates three output variables that constitute the final recommendation.

2.3.1. Overall Fitness

An aggregated score representing the user's overall fitness level. Range: [3, 9].

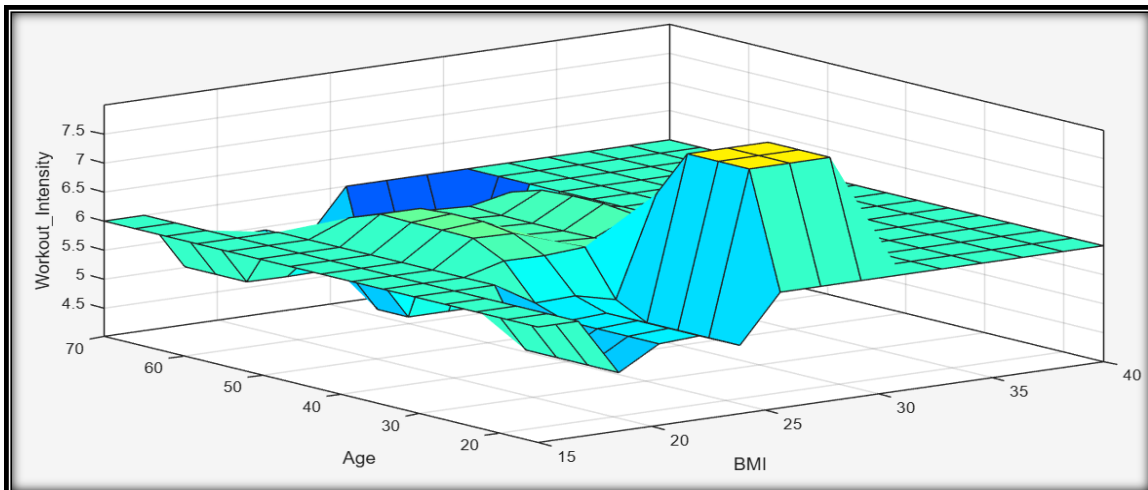
MF Name	Type	Parameters
Poor	trimf	[3, 4, 5]
Average	trimf	[5, 6, 7]
Excellent	trimf	[7, 8, 9]



2.3.2. Workout Intensity

The recommended intensity level for workouts. Range: [3, 9].

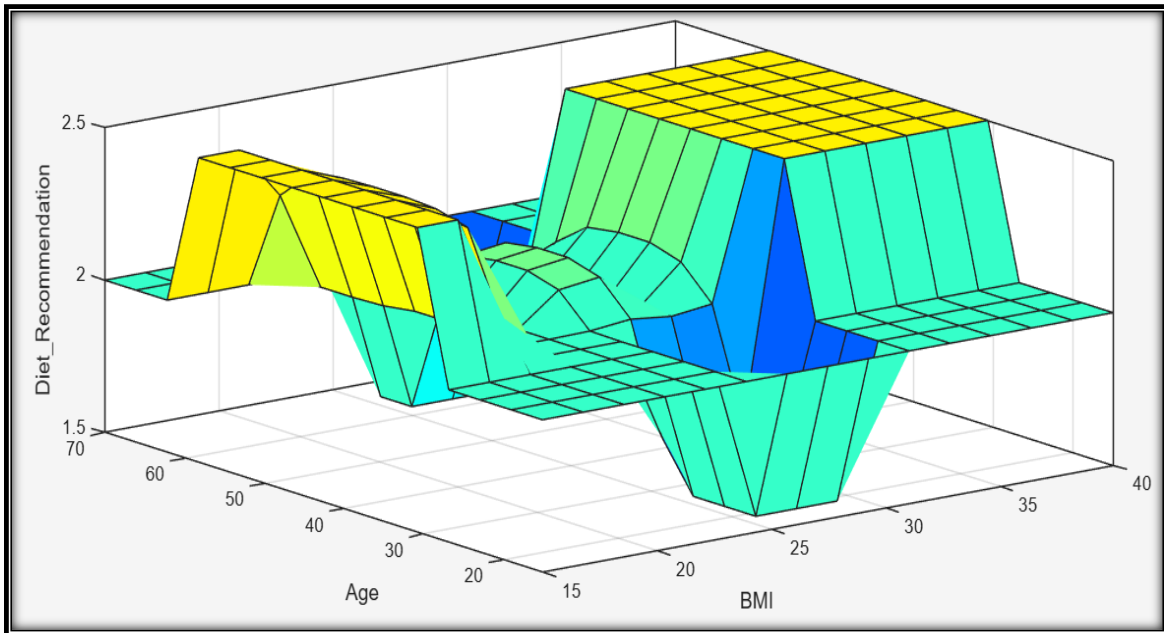
MF Name	Type	Parameters
Low	trimf	[3, 4, 5]
Moderate	trimf	[5, 6, 7]
High	trimf	[7, 8, 9]



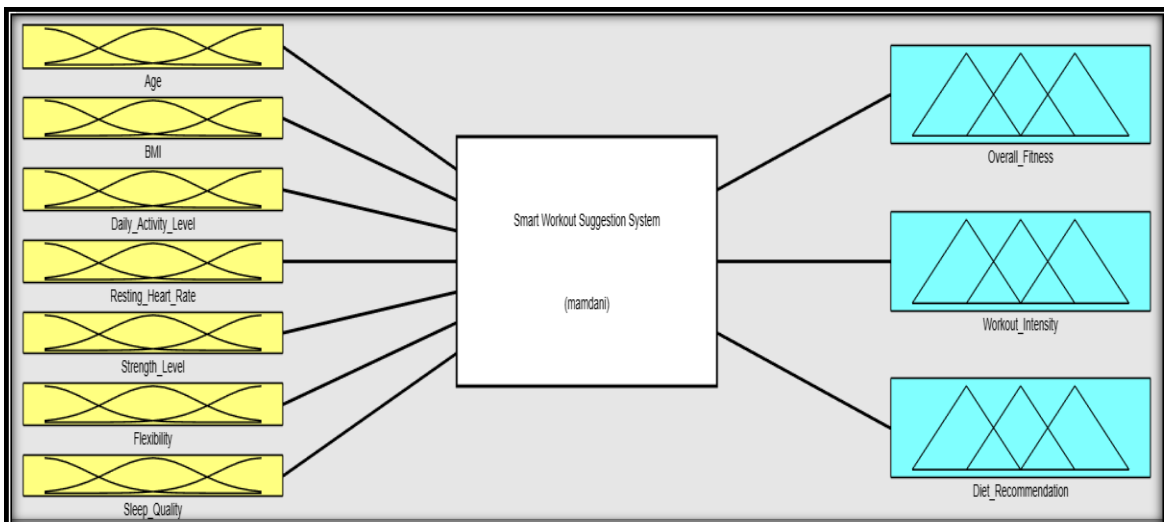
2.3.3. Diet Recommendation

A categorical recommendation for dietary focus. Range: [1, 3].

MF Name	Type	Parameters
Light Diet	trimf	[1, 1.5, 2]
Moderate Diet	trimf	[1.5, 2, 2.5]
High Protein Diet	trimf	[2, 2.5, 3]



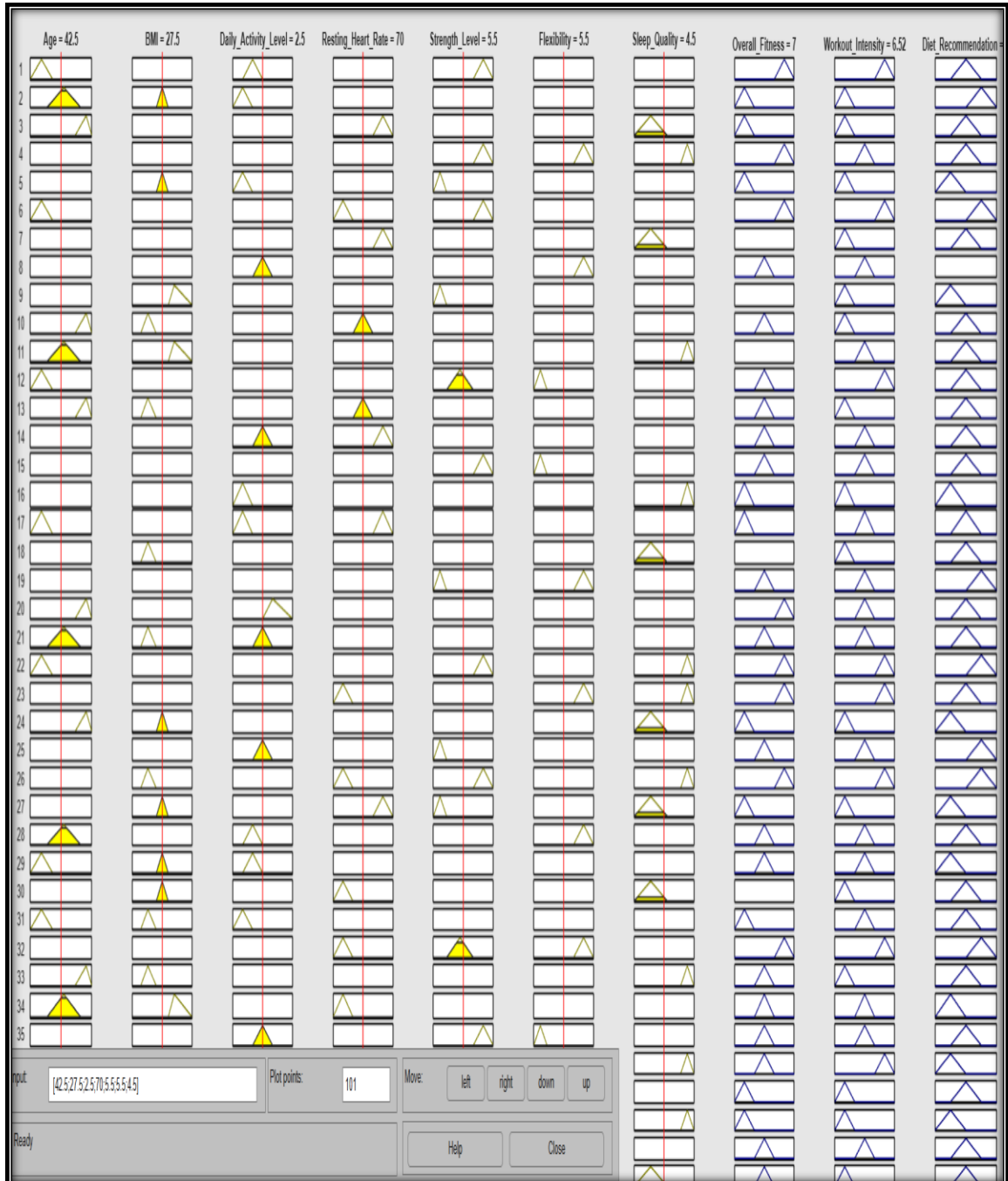
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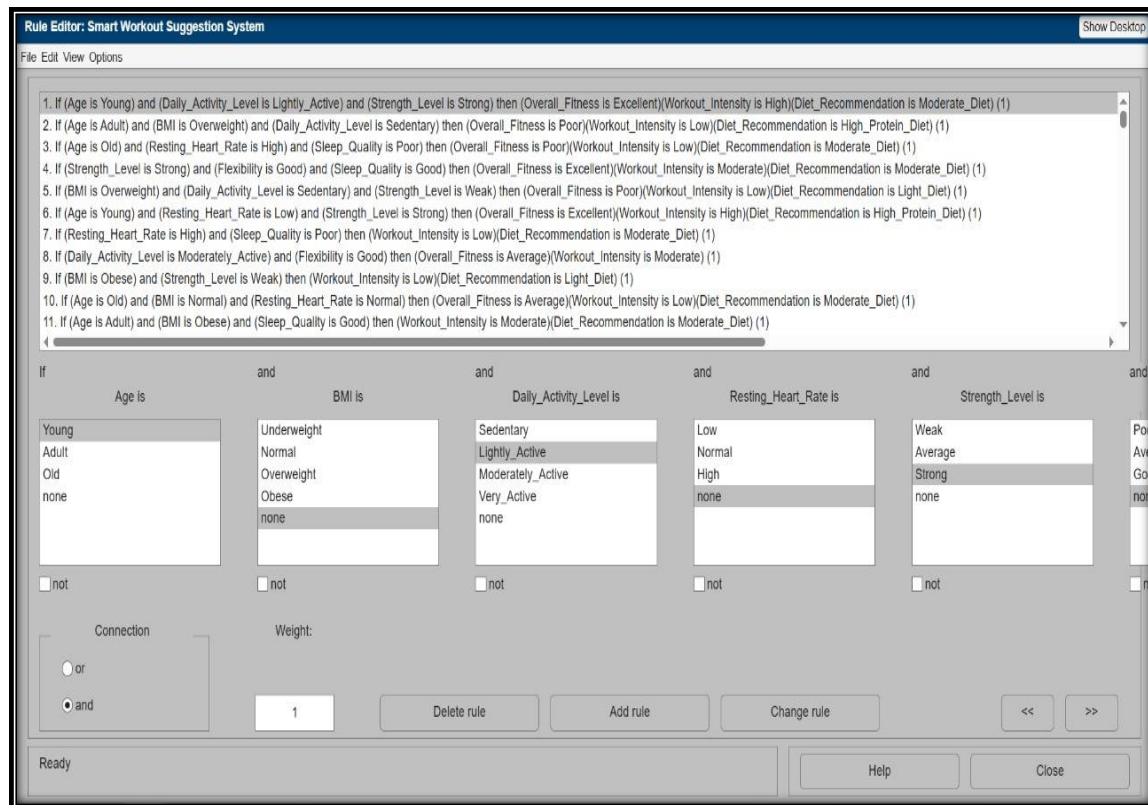


3. Rule Base

3.1. Rule Structure

The system's logic is encoded in 343 rules. Each rule follows a standard "IF-THEN" structure. The "IF" part, or antecedent, consists of one or more conditions on the input variables, connected by the logical AND operator. The "THEN" part, or consequent, specifies the resulting state of the output variables. All rules have a weight of 1.





It consists of 343 rules. Visit below to view all:

https://1drv.ms/u/c/a888350ebe303c80/IQAAOCsmHm3fS56d6GdM-tgFAZT3Nz_qSVtxVSXLML_3G0g?e=uwT7Zw

4. Results and Conclusion

4.1. System Impact and Benefits

The **Smart Workout Suggestion System** is engineered to function as a **virtual fitness consultant**, democratizing access to personalized health and wellness guidance. By leveraging a **sophisticated fuzzy logic framework**, the system translates **complex personal data into clear, actionable, and individualized recommendations**. This empowers users to make informed decisions about their fitness routines and dietary habits, removing the guesswork and common barriers associated with starting or optimizing a health journey.

The primary impact of this system is its ability to provide **holistic and adaptive advice**. It considers not just one or two metrics, but a **comprehensive profile of seven critical factors**—from physiological markers like **BMI** and **resting heart rate** to lifestyle elements like **daily activity level** and **sleep quality**. This **multi-dimensional analysis** ensures that

the recommendations are not only effective but also **safe and sustainable**. For example, the system will **intelligently lower workout intensity** for a user with poor sleep quality or high stress (indicated by a high resting heart rate), thereby **preventing overtraining** and reducing the risk of injury. This nuanced approach helps users build momentum, maintain motivation, and achieve **consistent progress**.

4.2. Conclusion

In conclusion, the **Smart Workout Suggestion System** represents a **significant step forward** in the application of **artificial intelligence to personal wellness**. It **bridges the gap between raw health data and practical, real-world application**, offering a level of **personalization previously available only through professional coaching**. By providing **scientifically-grounded and context-aware guidance** on workout intensity and diet, the system equips individuals with the tools they need to **take control of their health**, enhance their physical fitness, and ultimately **improve their overall quality of life**. It is a **powerful, accessible, and intelligent tool** designed to help people achieve their fitness goals **more efficiently and safely** than ever before.

THE END